

Asghar Afzal

Senior Frontend Architect · Senior Frontend Developer

asgharafzal479@gmail.com · +92 316 4343973 · Lahore, Pakistan · [linkedin.com/in/asghar-afzal](https://www.linkedin.com/in/asghar-afzal) · asgharafzal.com

PROFESSIONAL SUMMARY

Frontend Architect and Lead Engineer with 5+ years of end-to-end ownership over enterprise-grade systems — from architecture decisions to production deployments serving NHS hospitals across England. I define the stack, build the design system, set API contracts, lead the team, and own the outcome. Specialist in React/Next.js architecture, WCAG-compliant design systems, real-time AWS infrastructure, and AI-augmented development (Cursor, Claude, Copilot). Proven results: 40% reduction in UI inconsistency, 30% fewer production bugs, 35% faster delivery cycles through AI tooling.

CORE COMPETENCIES

Architecture & Design: Frontend Architecture, System Design, Scalability Patterns, API Contract Design, Micro-frontend

Frontend: React.js, Next.js, TypeScript, Vue.js, Redux, React Query, GraphQL, Storybook, Design Systems, WCAG 2.1

Performance: Core Web Vitals, SSR/SSG/ISR, Bundle Optimization, Lighthouse, Code Splitting, CDN, Lazy Loading

Styling: Tailwind CSS, Styled-Components, MUI, Ant Design, Design Tokens, Chromatic, npm Publishing

Backend & Cloud: Node.js, Express.js, GraphQL Server, WebSockets, AWS (EC2, S3, Lambda, Chime SDK), GCP, Docker, CI/CD

Databases: MongoDB / DocumentDB, MySQL, DynamoDB

AI Dev Tools: Cursor, Claude, GitHub Copilot — code generation, refactoring, architecture exploration

Testing & Leadership: Jest, RTL, PR Reviews, Sprint Planning, Agile/Scrum, Technical Mentorship

AI-AUGMENTED ENGINEERING & DEVELOPER PRODUCTIVITY

- ▶ Claude + Cursor: Architecture planning, generating test cases from requirements, technical documentation, and context-aware code generation — achieving 30–40% faster feature delivery on production projects.
- ▶ GitHub Copilot: Integrated into VS Code for autocomplete and boilerplate elimination
- ▶ Prompt Engineering: Reusable prompt templates for PR review summaries, architecture docs, and converting acceptance criteria into test specifications — ~35% less time on repetitive coding tasks.

PROFESSIONAL EXPERIENCE

Senior Frontend Architect / Lead Engineer · InnoSTARK Technologies · Lahore, Pakistan *Jan 2023 – Present*

- ▶ Owned end-to-end frontend architecture for NHS-affiliated platforms — stack selection, design system, and component infrastructure decisions across 3+ production portals serving thousands of NHS England patients.
- ▶ Architected and published a WCAG 2.1-compliant Design System (React + Storybook) to npm/Chromatic — reducing UI inconsistency by ~40% and enabling 60% faster design-to-code delivery.
- ▶ Designed real-time Video Consultation Platform replacing Jitsi with AWS Chime SDK — owned infrastructure decision, Node.js Lambda API layer, GraphQL schema. Zero patient-facing downtime during migration.
- ▶ Led Vue-to-React migration; defined component architecture and phasing strategy; enforced PR review culture and ADRs — cutting production bug rate by ~30%.
- ▶ Integrated Claude, and Copilot into team workflow — established AI code review standards achieving ~35% faster delivery. Mentored 5+ engineers on modern architecture and AI-assisted development.

Senior Frontend Developer · Techbucks · Lahore, Pakistan *Feb 2022 – Dec 2022*

- ▶ Led a team of 3 — owned technical direction, sprint planning, and PR review culture; delivered all projects on schedule with zero critical post-release issues.
- ▶ Designed reusable UI component library eliminating ~35% of repetitive dev effort; architected Node.js/Express.js + MongoDB API layer removing full-stack bottlenecks across concurrent projects.
- ▶ Managed GCP deployments and ran internal engineering workshops on React architecture, state management, and database design — raising the team's technical baseline.

Frontend Developer · Ursol · Lahore, Pakistan *Dec 2020 – Dec 2021*

- ▶ Delivered accessible, cross-browser React UIs backed by Node.js APIs; led frontend architecture for an AI-powered application from requirements through stakeholder demo delivery.
- ▶ Established responsive design patterns, component standards, and accessibility guidelines adopted as team-wide engineering baselines.

KEY PROJECTS

AttendAnywhere — NHS England Telemedicine Platform <https://england.nhs.attendanywhere.com/>

Stack: Next.js · AWS Chime SDK · Node.js · GraphQL · TypeScript · Storybook · AWS Lambda, .NET

- ▶ Full architecture ownership for a government-backed NHS telemedicine system. Made the infrastructure decision to replace Jitsi with AWS Chime SDK after evaluating reliability, HIPAA alignment, and latency — migrated with zero patient-facing downtime. Designed GraphQL schema, Lambda API layer, and WCAG 2.1-compliant Design System across all portals.

Zesty — Multi-Award-Winning NHS Patient Portal <https://www.inductionhealthcare.com/>

Stack: React.js · TypeScript · Storybook · Chromatic · Vue.js · Bootstrap · Ruby on Rails

- ▶ Architected WCAG-compliant Design System across multiple UK NHS hospital portals with Chromatic visual regression testing. Led Vue-to-React migration — defined component structure, data flow, and phasing strategy, improving maintainability and reducing average feature load time.

Skainet — AI Document & Workflow Platform <https://skainet.ai/>

Stack: *Next.js · Node.js · WebSockets · MongoDB · Redux · NLP · Styled-Components · Python*

- ▶ Full 0→1 architecture ownership: Next.js SSR, WebSockets for real-time doc sync, MongoDB for flexible document storage, Python NLP pipeline integration. Led team from blank repo to production.

RAAS — Crowdfunding & P2P Lending Platform

Stack: *Next.js · Node.js · Redux · MongoDB · Storybook · Tailwind CSS · Express.js*

- ▶ Defined frontend fintech architecture from scratch —. Built Storybook component library and integrated third-party financial services for end-to-end loan origination workflows.

Asset Planet — Asset Management Platform <https://assetplanet.com/>

Stack: *React.js · Redux · Node.js · Express.js · Ant Design*

- ▶ Refactored UI component architecture and introduced centralized API calling structure on a live platform — delivering measurable load time improvements and significantly reducing maintenance overhead.

Suntrix — Solar Energy Platform <https://suntrix.com.pk/>

Stack: *Next.js · SSR · GraphQL · React Query · MUI · Directus*

- ▶ Defined full tech stack and built application architecture from zero; leveraged Next.js SSR and GraphQL to achieve strong Core Web Vitals scores and first-page SEO rankings from launch.

GLT Dashboard — Shipping & Logistics ERP <https://gltmena.com/>

Stack: *Next.js · React.js · Node.js · Redux · Styled-Components · Express.js*

- ▶ Rebuilt ERP dashboard architecture end-to-end — redesigned state management, optimized API integration patterns, and led frontend team through full delivery of a complex multi-module logistics system.

EDUCATION

B.Sc. Software Engineering · COMSATS University Islamabad, Lahore Campus 2017 – 2021

Frontend Architect · React Architect · Lead Frontend Engineer · Next.js Architect · Design System Lead · NHS Healthcare · Performance Engineer · AI-Assisted Development · TypeScript Expert